MIL-S-44025
16 May 1980
(See 6.4)

MILITARY SPECIFICATION

SPECTACLES, LASER PROTECTIVE (WITH CASE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of laser protective spectacles.

1.2 Classification. Components shall be furnished in the following sizes, as specified (see 6.2).

Lens sizes: 44, 46 and 48 millimeters (mm)
Temple lengths: 6, 6-1/4, 6-1/2, 6-3/4 and 7 inches (152, 159, 165, 171 and 178 mm)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GGG-S-620	-	Spectacles, Industrial; And Lenses (Opthalmic, Plano And Prescription, Glare and Impact Resistant)
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes, Setup
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8465

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MIL-B-131 - Barrier Materials, Waterproof, Flexible, Heat Sealable
MIL-C-44026 - Case, Laser Protective Spectacles

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Z87.1-1979 - Practice for Occupational and Educational Eye and Face Protection

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York City, NY 10018.)

UNITED STATES DEPARTMENT OF COMMERCE

National Bureau of Standards NBS
Circular 533 or NBS Special
Publication 374 (cat. no. C13.10374)

(Application of copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Spectacles.

3.1.1 Spectacles furnished under this specification shall comply with the applicable requirements of ANSI Z87.1-1979 except as otherwise specified herein.

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3.2 Components and materials.

3.2.1 Components. Spectacles shall consist of plastic fronts, plastic side shields, plastic covered cable temples, lenses and appropriate hardware. Spectacles shall be furnished with headbands to insure firm fit to the face of the wearer.

3.2.2 Materials.

3.2.2.1 Plastic. The material used to make the plastic components, except the LASER-GARD for the side shields (see 3.2.2.2) and the tubing covering the temple cable section (see 3.2.2.3), shall be cellulose acetate or any other plastic having no toxic effects on the skin and having no offensive odors, and which will produce frames meeting all the requirements of 3.4.1.1. Except for the LASER-GARD side shields, the molded components shall be black in color with a dull finish.

3.2.2.2 LASER-GARD(R). The material for the side shields shall be Glendale Optical Company, Inc. (a subsidiary of American Cyanamid Company) LASER-GARD material formulation A, or equal having an optical density greater than 5.7 units at 694.3 nm (nanometer) and 12 units at 1060 nm when tested as specified in 4.3.1.1. The visual transmittance of the LASER-GARD shall be not less than 20 percent when tested as specified in 4.3.1.1.

3.2.2.3 Tubing. The covering for the cable section of the temples (see 3.4.2) shall be clear flexible tubing consisting of ethylene-vinyl acetate copolymer, by Thermoplastics Processors, Inc., Formulation #UE 630 or equal. The clear surgical tubing shall have a softening point greater than 150°F (65.6°C) with a Shore Type A Durometer hardness of 94 $\pm$ 5 (see 4.2 and 6.3).

3.2.2.4 Lens material and quality. Lenses shall be high quality absorptive glass.

3.2.2.5 Frame hardware. Hinges, rivets, eyelets, temple core and screws shall show no corrosion when tested as specified in 4.3.1.1.

3.2.2.6 Headbands. Headbands shall consist of black elastic webbing, 3/8 to 1/2 inch (10 to 13 mm) wide, and with appropriate hardware to allow adjustment of the band by the individual wearer (American Optical Corp., Allstar Glass-Gard Product Code No. 23536 or equal). The headband shall be attached to the ends of the temple cables.

3.3 Design, shape and sizes.

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3.3.1 Fronts. The fronts shall be designed to accommodate the shape lenses specified in 3.3.2 and shall have a universal fitting type bridge in the bridge sizes specified in 3.3.2.

3.3.2 Lenses and bridges. Lenses shall be plano and shall conform to shape S-7 in sizes and dimensions as specified in GGG-S-620. The lens sizes shall be 44, 46 and 48 mm as specified in GGG-S-620. The 44 and 46 lens sizes shall have a bridge of 22 ± 2 mm and the 48 lens size shall have a bridge size of 24 ± 2 mm.

3.4 Construction.

3.4.1 Frames. The lens frames, nose bridge, and end pieces of the fronts shall be made of one-piece plastic material. Frames shall have a dull finish, shall afford a reasonable degree of comfort to the wearer, and shall be readily fitted in the conventional manner. Lens frames shall contain grooves to fit accurately with lenses specified, and shall be provided with high-positioned end pieces designed to hold temple hinges. Temple hinges shall be riveted to the end pieces and temples in such a manner that they will not work loose or pull through the plastic. Temple hinges shall be of the five-barrel or the seven-barrel type, and shall pivot on a screw to provide for ready replacement of temples. Screws shall be clinched or otherwise secured to prevent backing off. The perpendicular distance from the center of the temple hinge to a line connecting the geometric center of the lens frame shall be not less than 6 mm. Construction of the lens frames shall be such that lenses may be removed and replaced by a qualified optometrist.

3.4.1.1 Strength of frames. Frames shall withstand the mechanical tests specified in 4.3.2.

3.4.2 Temples. The temples shall be of the metal cable type and shall be prefitted to the average ear shape. Temples shall be of such design as to permit ready adjustment and shall fit comfortably and securely on the wearer. The cable temples shall be permanently covered from the hinge to the cable portion with plastic material with dimensions sufficient to accommodate marking as specified in 3.6. The cable section shall be covered with flexible plastic tubing specified in 3.2.2.3. The terminal ends of the flexible plastic shall be secured to prevent any tendency to slip off the metal core. Cable temples shall be provided in 6, 6-1/4, 6-1/2, 6-3/4 and 7 ± 3/32 inch (152, 159, 165, 171 and 178 ± 2 mm) overall lengths.

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3.4.3 Side shields. Frames shall be equipped with two plastic side shields without perforations designed to prevent entry of direct unfiltered light (see 3.2.2.2). The side shields shall have an integrally molded bead to obtain the necessary rigidity. Side shields shall be pivoted to permit their folding for compactness when spectacles are not in use. Side shields shall be 1-1/8 ± 1/8 inches (29 ± 3 mm) deep at the deepest part and shall be tapered with an anatomic periphery extending at least halfway around the circumference of the lens frame. Side shields shall not be easily detachable from the frame. Snap-on or clip-on types of side shields shall not be acceptable. For wearing, the side shields shall be positioned against the frame so that no direct light is visible to the wearer between the frame and the side shield. The shields shall remain in this position during normal handling.

3.4.4 Lenses.

3.4.4.1 Matching. Within the visual center of each lens, the right and left lenses of each pair of spectacles shall be matched in sign and prismatic power to within 0.125 prism diopter in the vertical direction.

3.4.4.2 Lens defects. Lenses shall be free from striae, waves, bubbles or any other defects which may impair their optical quality when tested as specified in 4.3.4. Optical distorsion within 3 mm of the edge of a lens is acceptable. Edges of the lenses shall be smooth and dull finished.

3.4.4.2.1 Definition. Lenses shall show clear definition when tested as specified in 4.3.1.1.

3.4.4.3 Prismatic power. The prismatic power shall not exceed 0.125 prism diopter when tested as specified in 4.3.1.1.

3.4.4.4 Refractive power. Lenses shall have refractive power not greater than plus or minus 0.125 diopter in any meridian, and the difference in refractive power between any two meridians shall not be greater than 0.125 diopter when tested as specified in 4.3.1.1.

3.4.4.5 Curvature. The curvature of the lenses shall be 6.00 ± 0.12 diopters, when tested as specified in 4.3.1.1.

3.4.4.6 Impact. Lenses shall be heat-treated or chemically tempered so as to withstand impact specified in 4.3.1.1.

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3.4.4.7 Optical density. The spectacles shall have an optical density of not less than 5.9 units at 694.3 nm (nanometer) and no less than 12 units at 1060 nm when tested as specified in 4.3.1.1.

3.4.4.8 Visual transmittance. The photopic visual transmittance of the lenses shall be a minimum of 32 percent and additionally shall have a transmittance greater than 35 percent at 500 nm when tested as specified in 4.3.1.1.

3.4.4.9 Lens damage. The lenses shall not be damaged by the stresses resulting from the incidence of 4 joules per square centimeter and an average power density of 150 megawatts per square centimeter when tested as specified 4.3.1.1.

3.4.5 Complete spectacles.

3.4.5.1 Lens fit. Lenses shall fit the frames with no evidence of spaces existing between the frames and lenses when tested as specified in 4.3.4.

3.4.5.2 Surface quality. The lens surfaces shall be of a quality such that they appear black against a black background with side illumination when tested as specified in 4.3.4.

3.5 Carrying case. A carrying case conforming to MIL-C-44026 shall be furnished with each pair of laser protective spectacles.

3.6 Marking.

3.6.1 Frames. All frames shall bear a permanent lens size designation (see 1.2), and shall have a distinctive mark by which the manufacturer may be readily identified.

3.6.1.1 Cable temples. Cable temples shall be marked in the following manner:

The left outside cable temple shall have embossed in letters 0.12 inch min. (3 mm min.) high the following legend: "LASER PROTECTIVE".

The right outside cable temple shall have embossed in letters 0.12 inch min. (3 mm min.) high the following legend: "OD6 RUBY-OD12-ND".

The right and left cable temples shall have embossed on the inside, in letters .06 $\pm$ .02 inch (1.5 $\pm$ 0.5 mm) high the name and/or symbol of the manufacturer and the length of the cable temple (see 1.2).

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3.6.2 Lenses. All lenses shall bear a neat, permanent, distinctive mark by which the manufacturer may be readily identified. All markings shall be near the edge, out of the field of vision.

3.6.3 Information sheet. Each carrying case shall have inserted into it sheet of white paper with the following clearly and neatly printed information:

"Laser Protective Spectacles

These laser protective spectacles have been designed to protect against possible harmful effects of specific wave lengths of laser light. In order to provide the best possible protection, it is necessary that the spectacles be adjusted correctly by personnel trained to perform this task--optometry officer or optical laboratory specialist.

A. Instructions to the User.

1. Fitting: Spectacles must fit snugly for maximum protection. When properly fitted the side shields should bear against the face. Whenever the glasses are not in proper adjustment, return to the Optometry Clinic for re-fitting. Do not attempt to adjust the spectacles yourself. Always have the spectacles adjusted by Medical Department personnel trained to perform this function.

2. Care of spectacles: The lenses will scratch if they are not cared for properly. Always keep the spectacles in their case when not being used. To clean the lenses, wash off all dust and dirt with water or a liquid cleaning solution and dry with a soft tissue or cloth. If spectacles are damaged or do not fit properly, return them to the Optometry Clinic for repair or replacement.

3. The following points are important for your safety:

a. These spectacles are for use with Ruby, Gallium-Arsenide, and Neodymium Lasers ONLY.

b. Wear the spectacles at all times when you are in the area where these laser devices are being utilized.

c. Do not look directly into laser beam.

d. You will not be able to see the color red (warning lights, signal smoke, etc.) while wearing Laser Protective Spectacles.

e. Laser Protective Spectacles are not sunglasses.

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B. Instructions to Fitter: The fitting and adjustment of Laser Protective Spectacles is critical to the safety of the user. The following guidance is provided to help you in this task:

1. The spectacles are designed with a universal fit bridge (saddle type) and come in three sizes:

a. 44-22
b. 46-22
c. 48-24

2. Select the proper size according to the width of the patients head to minimize the adjustments required on the side shields.

3. Adjust the spectacles so they are as comfortable as possible while at the same time they fit snugly against the face to preclude light leakage.

a. If side shields press too tightly against the face (assuming proper frame size has been selected) file off the back edge of shields until they fit comfortably.

b. If side shields do not touch the face, heat the bridge and bend to give more faceform to the spectacles.

c. Adjust the comfort cable temples so they conform to the anatomy of the patient's ears and still hold the spectacles snugly against the face.

d. Check the fit by shining a penlight around the periphery of the frame to determine if any light leaks exist.

4. The universal fit bridge and the three sizes should allow an adequate fit to almost all personnel.

5. The fitter should carefully examine the spectacles for any damage to the lenses that may affect their ability to provide adequate protection (i.e., cracks, deep scratches, etc.). This is especially important with spectacles being reissued to a new individual and during subsequent readjustment vists.

6. When temples require placement, they must be replaced only with temples designed for this type laser spectacles, since the temples contain the information concerning the protection afforded by the lenses."

3.7 Interchangeability. All parts of the same manufacturer's model shall be constructed to definite standards and tolerances in order that such parts may be interchanged, replaced, or adjusted without modification.

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3.8 Workmanship. The finished spectacles shall conform to the quality of product established by this specification and the occurence of defects shall not exceed the specified acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification. A certificate of compliance shall be furnished to verify plastic tubing requirements specified in 3.2.2.3.

4.3 Quality conformance inspection. Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise specified hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase document.

4.3.1.1 Testing of components. The components and materials shall be tested for the applicable characteristics in table I. The lot size shall be expressed in the same units as the sample unit. The inspection level shall be S-1 for all tests (see 4.3.3.1.2 for lens size option) except prismatic power, refractive power, definition and impact resistance. For prismatic power, refractive power, definition and impact resistance, inspection shall be on a 100 percent basis. The sample unit for the metal parts shall be one part. The sample unit for the lenses shall be one lens. The sample unit for the LASER-GARD shall be one part. Failure of any sample unit shall be cause for rejection of the lot.

4.3.2 In-process testing. Testing of the partially fabricated spectacles shall be performed in accordance with table II for the characteristics shown therein. The lot size shall be the number of spectacles in the production lot. The sample unit for the matching test shall be one front with lenses in place. For the matching test there shall be 100 percent inspection, except for Government verification testing the inspection level shall be S-1. The sample unit for the mechanical tests shall be 3 fronts (2 with lenses and 1 without lenses) and 1 frame without side shields or lenses. For mechanical tests, the inspection level shall be S-1. Failure of any sample unit shall be cause for rejection of the lot. One of the fronts with lenses should be selected prior to attaching the hinges for use in lens container impact test.

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INSTRUCTIONS FOR TESTING

TABLE I

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Metal components:									
Corrosion	3.2.2.5	4.4.9		X	1		X	-	
Lens material and quality	3.2.2.4	1/		X	-		X	-	
LASER-GARD:									
Optical density	3.2.2.2	4.4.6 1/ 6/		X	1		-	0.1 unit	
Visual transmittance	3.2.2.2	4.4.7 1/ 6/		X	1		-	1 percent	
Lens 7/:									
Prismatic power	3.4.4.3	3/		X	1		X	- 4/	
Refractive power	3.4.4.4	3/		X	1		X	- 4/	
Curvature	3.4.4.5	4.4.5		X	1		X	-	
Impact	3.4.4.6	2/		X	1		X	- 4/	
Definition	3.4.4.2.1	5/		X	1		X	- 4/	
Optical density	3.4.4.7	4.4.6		X	1		-	0.1 unit	
Visual transmittance	3.4.4.8	4.4.7		X	1		-	1 percent	
Lens damage	3.4.4.9	4.4.8		X	1		X	-	

1/ The contractor shall submit a certificate of compliance indicating conformance to the required characteristics.

2/ Impact test in accordance with GGG-S-620.

3/ Shall be tested in accordance with applicable test method of ANSI Z87.1 except calibration and measurements shall be to 1/8 diopter.

4/ For this test, 100 percent inspection shall be used except for Government verification testing. For Government verification testing, the inspection level shall be S-2.

5/ The lenses shall be tested for definition using the telescope specified in 4.4.1 with the NBS Special Publication 374 or NBS Circular 533 high contrast chart. The number 20 vertical and horizontal line patterns on the chart shall be clearly resolved at the distance of 35 feet (10.668 m) with a telescope focus adjustment not to exceed ± 1/8 diopter.

6/ The optical density and the visual transmittance shall be measured at the thinner center position of the side shield.

7/ All of these tests should be carried out on each melt. In the absence of melt designation by the contractor, all of these tests shall be carried out on each shipment of glass.

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TABLE II. In-process tests

Characteristic	Requirement	Test method	Individual unit	No. deter. per sample unit	Results reported as pass or fail
Matching	3.4.4.1	4.4.1	X	1 each pair lenses	X
Mechanical tests:					
Aging	3.4.1.1	1/	X	1	X
Flat transverse	3.4.1.1	1/	X	1	X
Edge transverse	3.4.1.1	1/	X	1	X
Nose-bridge bending	3.4.1.1	1/	X	1	X
Shape retention	3.4.1.1	1/	X	1	X
Impact, lens containers	3.4.1.1	1/	X	1	X
Hinge	3.4.1.1	1/	X	1	X

1/ Shall be tested in accordance with the applicable test method in GGG-S-620.

4.3.3 Inspection of the end item.

4.3.3.1 Examination of the end item. The end item shall be examined for visual and dimensional defects in accordance with the classification of defects set forth in 4.3.3.1.1 through 4.3.3.1.2. The inspection levels and acceptable quality levels (AQLs) shall be as specified in 4.3.3.1.3. The lot size shall be expressed in units of spectacles with cases for examination in 4.3.3.1.1 and 4.3.3.1.2.

4.3.3.1.1 Visual examination. The sample unit for this examination shall be one pair of spectacles with case.

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Examine	Defect	Classification Major	Minor
Lens	Bubble or seed greater than 0.2 mm, hairline scratch, striae or other imperfection at the visual center of the lens.	X	
	Bubble, seed greater than 0.5 mm, or striae outside the visual center of the lens.		X
	NOTE: The visual center of the lens shall be defined as a circle, 24 mm in diameter, centered in the lens.		
	Hole.	X	
	Any chip more than 0.016 inch (0.4 mm) from the edge of the lens.	X	
	Any chip extending more than .063 inch (1.6 mm) along the edge of the lens.	X	
	Chip .016 inch (0.4 mm) or less from the edge of the lens.		X
	Chip extending .063 (1.6 mm) or less along the edge of the lens.		X
	Three or more chips .016 inch (0.4 mm) or less from edge or extending less than .063 inch (1.6 mm) along the edge of lens.	X	
	Any hairline scratch (outside the visual center) more than 0.125 inch (3.2 mm) long.		X
	Three or more hairline scratches (outside the visual center).	X	
Frame (including front, temples and side shields)	Not plastic (except cable portion of temple).	X	
	Not dull finish (except for the plastic tubing).		X
	Not clean.		X
	Not smooth.		X
	Not black (except for the side shields and the plastic tubing).	X	

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Examine	Defect	Classification Major	Minor
Design	Any characteristic not accomplished as specified.	X	
	Not S-7 shape (lens and frame).	X	
	Bridge not universal type.	X	
	Not provided with high positioned end pieces.	X	
Construction and workmanship	Component missing, broken or cracked.	X	
	Lens edges not smooth.	X	
	Component bent out of shape, twisted, malformed or otherwise damaged.	(*)	
	Hinges improperly riveted to end pieces.	(*)	
	Temple hinges not of the five-barrel or the seven-barrel type.	X	
	Temple hinges not pivoted on a screw for easy placement of temples.	X	
	Any burr or sharp edge than may cause injury.	X	
	Cable temples not adjustable to fit comfortably and securely on the wearer.	X	
	Cable temple not prefitted to average ear shape.	X	
	Cable section not covered with clear flexible plastic tubing.	X	
	Cable section not secured.	X	
	Side shield has no integrally molded bead.	X	
	Side shields do not fold properly.	X	
	Side shields easily detachable.	X	
	Side shields snap or clip-on type.	X	
	Side shield does not move into position for wearing, or does not maintain that position in normal handling.	X	
	Any light visible between the frame and side shield when spectacles are being worn.	X	
Screws	Loose or not fully secured; unthreads easily when turned with a jewelers screwdriver.	X	
Assembly	Not connected or joined as specified, or assembly is poorly accomplished.	X	

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Examine	Defect	Classification Major	Minor
Headband	Not as specified.	X	
Carrying case	Missing.	X	
	Not clean.		X
	Dirt, oil, foreign matter, spot, stain or other discoloration.	(*)	
Marking	Missing, misspelled, illegible, incorrect.		X

(*) Defects so designated shall be classified as major when seriously affecting serviceability or appearance and minor when affecting serviceability or appearance but not seriously.

4.3.3.1.2 Dimensional examination. The sample unit for this examination shall be one pair of spectacles. Any dimension that is not within the specified tolerance shall be classified as a defect. The following dimensions are applicable to this examination:

Lens size (see 3.3.2) (As an option this may be done under 4.3.1.1)
Bridge size distance between lenses (see 3.3.2)
Perpendicular distance from center of temple hinge to a line connecting the geometric center of the lens, less than 6 mm (see 3.4.1)
Cable temple length (see 3.4.2)
Depth of side shields (see 3.4.3)

4.3.3.1.3 Inspection levels and AQLs for examinations. The inspection levels, for purposes of determining the sample size, and the AQLs expressed in defects per hundred units shall be as follows:

Examination paragraph	Inspection level	AQLs Major	Total
4.3.3.1.1	II	1.0	6.5
4.3.3.1.2	S-3	(one class)	4.0

4.3.3.2 End item testing. The end item shall be tested for the applicable characteristics in table III. Inspection shall be done on a 100 percent basis. Failure of any sample unit shall be cause for rejection of that unit. For Government verification testing, the inspection level shall be S-1.

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4.3.4 Packaging inspection. An examination shall be made to determine that the preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged except that it need not be closed. Defects of closure shall be examined on shipping containers fully packaged. The lot size shall be the number of containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Material	Any nonconforming component; component missing, damaged or otherwise defective. Evidence of barrier material delaminating. Color of barrier material not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, improper taping or inadequate stapling. Bulged or distorted container. Barrier material bags with heat sealed seams that are crooked, non-continuous, open or not parallel. Tear nicks or notches missing. Bag not style specified.
Content	Number per shipping container is more or less than required. Number of spectacles with headband per interior box is not as specified.

TABLE III. End item tests

Characteristic	Requirement paragraph	Test method	Individual unit	No. determinations per sample unit	Results reported as pass or fail
Lens removal	3.4.1	1/	-	-	-
Lens defects	3.4.4.2	4.4.2	X	1 each lens	X
Lens fit	3.4.5.1	4.4.4	X	1 each lens	X
Surface quality	3.4.5.2	4.4.3	X	1 each lens	X

1/ A certificate of compliance shall be acceptable for this requirement.

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4.4 Methods of inspection.

4.4.1 Matching. Matching of lenses for sign and prismatic power shall be determined with an 8 power telescope equipped with a 0.75 inch (19 mm) objective operture. The telescope shall be positioned 35 feet (10.7 m) from a diopter chart. The measurement of the prismatic power of both the left and right lenses shall be made with the lenses in their frame. The telescope objective shall be 15 mm to 35 mm from the outer surface.

4.4.2 Lens defects. Surface waves, striae, and other irregularities affecting the optical qualities of the lens shall be observed by moving the lens in front of the aperture of a lensometer instrument, or equal, with the horizontal axis of the lens being maintained parallel to the horizontal meridian of the instrument target in such a way that the target is observed successively through all parts of the lens. Sudden jumps or blurring of the image will indicate lens defects.

4.4.3 Surface quality. Lenses shall be tested for surface quality by examining them by side illumination of not less than 100-foot candles against a black background. A lens with satisfactory quality will appear black.

4.4.4 Lens fit. To test for lens fit, each pair of assembled spectacles shall be viewed in front of a strong light to assure that no spaces exist between the frames and lenses.

4.4.5 Curvature. The outside curvature of lenses shall be measured by a Geneva lens measure, or equal.

4.4.6 Optical density. Optical density of the lenses and of the side shield shall be measured on an appropriate spectrophotometer. A recording spectrophotometer shall be used to record the optical density (from 350 nm to 1060 nm) of each test specimen. The data to be obtained shall consist of the optical density measured at 694.3 nm and 1060 nm and plots of the optical density from 350 to 1060 nm. The lens or side shield may be ground to a thickness which will give suitable optical density measurements and the results extrapolated back to the original lens thickness. Alternatively, the optical density can be calulated using the method in 4.4.8.

4.4.7 Visual transmittance. The visual transmittance shall be measured with a CIE source C which approximates average daylight. A type C source is a gas-filled tungsten lamp calibrated at 2854 K having an appropriate set of filters. The combination of lamp and filters shall have a correlated color temperature of approximately 6500 K. A detector, with cell response matched to the standard luminosity curve of the eye, shall be used to measure illuminance with and without the spectacles. Transmittance shall be computed by dividing the illuminance measured with the spectacles by the illuminance measured without the spectacles. The transmittance at 500 nm shall be obtained from the spectrophotometer plots in 4.4.6.

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4.4.8 Lens damage. A Q-switched ruby laser shall be used to irradiate the spectacles with an average power density of 150 megawatts per square centimeter. An arrangement similar to that shown in figure 1 shall be used. The reference detector (detector 1) shall be calibrated to monitor the laser power output. A set of 2 to 6 optical density glass neutral filters shall be used to check the calibration of the system. The optical density of the spectacles shall be determined from the following equation:

$$\text{Optical density} = \log_{10} \frac{1}{T}$$

$$\text{Where T (transmission)} = K \frac{(V2)}{V1}$$

and K is a system constant determined by the ratio of V1/V2 with no sample in the test arragnement of figure 1.

The value of K shall be determined for the test setup used and the outputs of the detectors (V1 and V2) shall be recorded for each LASER firing. The LASER beam diameter shall also be recorded.

1/ Angle shall be greater than 30°.

Damage to the test item shall be regarded as any physical damage to the spectacle lenses such as bubbles, cracks, fogging or melting, and any temporary or permanent reduction of optical density below 5.9 for wavelength and power or energy densities below the minimum levels of 3.4.4.9. The required data shall be measured to the following accuracies:

a. Optical density, ± 0.1.
b. Transmission, ± 5%.
c. Energy density, ± 10%.
d. Power density, ± 10%.

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4.4.9 Corrosion-resistance test for metal parts. Metal parts shall be tested for corrosion resistance by placing them in a boiling 10-percent (by weight) aqueous solution of sodium chloride for a period of 15 minutes. The parts, upon being removed from the solution, shall be immediately immersed in a 10-percent (by weight) aqueous solution of sodium chloride at room temperature. The parts shall then be moved from this solution, and without having the adhering liquid wiped off, allowed to dry for 24 hours at room temperature. The metal parts shall then be rinsed in lukewarm water and allowed to dry. On examination, the metal shall show no signs of corrosive action.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A.

5.1.1.1 Barrier material bag. Each pair of spectacles with headband (see 3.2.2.6) shall be inclosed in a flat style heat sealed barrier material bag having inside dimensions of 6-3/4 inches (171 mm) in length by 3-1/2 inches (89 mm) in width (± 1/8 inch (3 mm) in each dimension). The bag shall be formed and closed with 3/8 inch (10 mm) (+ 1/16 inch (2 mm)) wide heat seals that are straight, continuous, and parallel to each other and the formed edges of the bag. A nick or notch between 1/16 inch (2 mm) to 1/8 inch (3 mm) deep shall be made on each side seam approximately 1 inch (25 mm) from one end of the bag. The bag material shall be fabricated by hot-extruding or laminating not less than 0.001 inch (0.03 mm) thick polyethylene on 0.00035 inch (0.009 mm) thick aluminum foil (± 10 percent) and laminating the plain side of the foil to 0.0005 inch (0.013 mm) thick polyester (± 10 percent). The polyester shall be bonded to the aluminum foil by hot-extruding 7 pounds (3.15 kg) per ream (± 10 percent) of polyethylene between the layers. The exterior surface of the bag material shall be colored olive drab.

5.1.1.1.1 Alternate bag material. An alternate bag material may be used, but it shall conform to class 1 of MIL-B-131. When used, the exterior surface of the bag material shall be colored olive drab.

5.1.1.2 Unit pack. One pair of bagged spectacles with headband shall be suitably positioned within a closed carrying case (see 3.5) that includes an anti-fogging kit and folded information sheet (see 3.6.3) and shall be placed in a folding paperboard box conforming to variety 1, style III, type G, class 1 of PPP-B-566 or in a setup paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676. Inside dimensions of each paperboard box shall approximate 7-1/2 inches (191 mm) in length, 4-1/4 inches (108 mm) in width, and 2-5/8 inches (67 mm) in depth. Approximate dimensions are furnished as a guide only. Box closure shall be secured with 2 inch (51 mm) minimum width tape conforming to type III, grade B of PPP-T-45, applied at the center of the length opening and along the bottom and up each side at least 1-1/2 inches (38 mm).

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. One hundred pair of spectacles of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed flat, five in length, two in width, and ten in depth within a shipping container. Inside dimensions of each shipping container shall approximate 22 inches in length, 15-1/2 inches in width, and 27 inches in depth (572 by 394 by 686 mm). Approximate dimensions are furnished as a guide only. Each shipping container shall be closed method III, waterproofed method V, and reinforced in accordance with the appendix of the container specification, except that the inspection shall be in accordance with 4.3.4.

5.2.2 Level B packing. One hundred pair of spectacles of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, five in length, two in width, and ten in depth within a shipping container. Inside dimensions of each shipping container shall approximate 22-1/2 inches in length, 15-1/2 inches in width, and 27 inches in depth (572 by 394 by 686 mm). Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification, except that the inspection shall be in accordance with 4.3.3.1.3.

5.2.2.1 Weather resistant containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed method III in accordance with the appendix of the container specification, excpet that the inspection shall be in accordance with 4.3.3.1.3.

5.2.3 Commercial packing. Spectacles, preserved-packaged as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

5.3.1 Special marking.

5.3.1.1 Barrier material bag. Sealed barrier material bags shall be marked on the outside surface with lens size and temple length for the spectacles conformed within the bag.

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5.3.1.2 Shipping container. Each shipping container shall be legibly printed, stenciled, or stamped in black ink with capital letters not less than 1/2 inch (13 mm) in height with the following precautionary marking:

STORE IN COOL, DRY AREA --

HIGH HUMIDITY AND TEMPERATURE

ARE DETRIMENTAL TO THE LENSES

Each unit pack shall have applied the same special marking as above, except that the capital letter size shall be in proportion to the unit pack size.

6. NOTES

6.1 Intended use. Laser protective spectacles are intended to provide an interim measure of protection against eye damage occoasioned by the tactical use of lasers. The spectacles will provide protection against lasers, particularly against ruby or neodymium lasers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of lenses (see 1.2).
(c) Length of temples (see 1.2).
(d) Selection of applicable level of packing (see 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Material (see 3.2.2.3) meeting this requirement is available from L.Frank Markel and Sons, Inc., Norristown, PA. It is identified as No. 7391.

6.4 Supersession. This specification supersedes LP/P. DES 18-76 dated 8 November 1976 for Spectacles, Laser Protective (With Case).

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors and symbols specified in ASTM E 380 Standard for Metric Practice and are for information only. In each instance, the value stated in US customary units shall be controlling.

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